MicroPulse

Quadrant View — Coach Guide

External vs internal load — Gabbett 2016 / 2017

FOR LITE CATAPULT TEAMS

1. Overview — what is Quadrant View?

Quadrant View (/coach/quadrant) is the page where MicroPulse tells you the Internal:External coupling story for the whole squad in a single chart. It plots every active player on a 2 × 2 grid — external load on the X-axis, internal cost on the Y-axis — and lets you see at a glance who is at injury risk, who is decoupled (early-warning fatigue), who is at peak fitness, and who is under-stimulated.

The framework comes from **Gabbett 2016 + 2017** — two of the most cited papers in modern team-sport load monitoring. The 2 × 2 quadrant idea originated in Vanrenterghem 2017 and was operationalised by Gabbett. The same chart structure has been used by elite organisations across football, rugby, AFL, and basketball.

Lite tier verdict

The 2 × 2 quadrant chart itself works at FULL fidelity on Lite — it is built on total_distance (external) and sRPE × duration (internal), both available on every Catapult plan. The Squad Load table below the chart is the part that changes: on Lite it swaps the Pro-only B2-3 efforts columns out for HSR / Sprint / Max Velocity columns. The page automatically detects your tier and serves the right variant.

Where does Quadrant View fit?

- Page: /coach/quadrant — accessible from the Monitoring section in the sidebar.
- Visible to all Catapult plans (Lite and Pro / Premium). The page adapts to your tier automatically.
- Citation badge at the top tells you which variant is active: "Gabbett 2016 · Volume axis" on Lite (blue badge), "Gabbett 2017 · Full axis" on Pro / Premium (green badge).

2. The chart — what the two axes mean

The chart has two universal inputs that work on every Catapult plan, including Lite. Understanding what they measure is the first step to reading the chart correctly.

Axis	What it measures	Where the number comes from
X-axis — External load	Average total distance per day across the selected window (7 / 14 / 28 days). Measured in metres per day.	Catapult total_distance — available on every plan. Sum across the window ÷ days with data.
Y-axis — Internal cost	Average sRPE × duration per day across the selected window. Measured in AU (arbitrary units) per	Player-logged session RPE × session duration. Captured via the player app after every session.

Axis	What it measures	Where the number comes from
	day.	

The two inputs are intentionally orthogonal. **External load** tells you what the body did. **Internal cost** tells you how hard the body felt it. The relationship between the two is the whole point — when external is high but internal is low (peak fitness) or external is low but internal is high (decoupled / fatigued), that's where the actionable insight lives.

Quadrant lines — at the team median

The lines that divide the chart into four quadrants are NOT fixed thresholds. They are placed at the team median for both axes — half your players land left of the X-line, half land right; same for Y. That means quadrant membership is RELATIVE to the rest of your squad on a given day. A player can move between quadrants from week to week even if their absolute load doesn't change much, depending on what their teammates are doing.

Dot size and colour

- Dot SIZE reflects ACWR (acute:chronic ratio) — larger dot = higher ACWR. ACWR ≥ 1.3 makes the dot noticeably larger.
- Dot COLOUR reflects ACWR severity — RED if ACWR ≥ 1.5, AMBER if 1.3 — 1.5, GREEN otherwise.
- ACWR is computed from internal cost (acute 7-day vs chronic 28-day) — it works fully on Lite because it's RPE-driven, not GPS-driven.

3. The four quadrants — what each means

Read the chart by quadrant first, then by individual dot. Here is what each quadrant tells you and what to do about it.

Quadrant	What it means	What you should do
INJURY RISK (top-right)	Player runs a lot AND pays a high internal cost. They are working hard and their body is registering it. This is the highest-risk position — chronic accumulation of high external + high internal load is what Gabbett's data linked to soft-tissue injury.	Reduce load in the next session. If multiple players are clustered here, the team's last 7-14 days have been too heavy — pull intensity back across the squad.
DECOUPLED (top-left)	Player runs LITTLE but pays a HIGH internal cost. Classic early-warning fatigue or illness signal (Halson 2014). Body is working harder than the work demands — a sign of poor recovery, sleep debt, or sub-clinical illness.	Talk to the player. Check Decision Summary wellness scores. Consider a sleep / RHR check. This is often the first signal before a flu / overtraining episode.
PEAK FITNESS (bottom-right)	Player runs a lot but pays a LOW internal cost. Well trained, fit and fresh. The body is doing the work efficiently. This is exactly where you want most of your squad to sit during peak parts of the season.	No specific action — keep the work going. These players can handle additional load if needed.
UNDER-STIMULATED (bottom-left)	Player runs little, pays little internally. No injury risk — but also possibly not getting enough	Check whether the under-loading is intentional (RTP plan)

Quadrant	What it means	What you should do
	stimulus to build / maintain fitness. Common in returning-from-injury cases or players with reduced training minutes.	or accidental (missed sessions). If accidental, plan a top-up. If intentional, just monitor — they'll move out of this quadrant when the plan calls for it.

4. The quadrant counts bar

Above the chart you'll see four count cards — one per quadrant — showing how many players are in each. This is the 5-second team-state read.

How to read the counts

Multiple players in INJURY RISK: team has been over-loaded this week. Pull back team-wide intensity in the next session.

Multiple players in DECOUPLED: team is fatigued. Could be a virus going through the dressing room, a heavy travel block, or accumulating chronic load. Investigate the cluster.

Most players in PEAK FITNESS: team is in great shape. Good time for high-intensity competitive sessions or matches.

Most players in UNDER-STIMULATED: training volume is too low for the squad. Common in early pre-season or after long international breaks. Plan a build-up block.

5. The time window — 7 / 14 / 28 days

The buttons in the top-right (7d / 14d / 28d) change the window over which the chart aggregates external + internal load. Each window answers a different question.

Window	What it shows	When to use
7 days (default)	Current micro-cycle. Last 7 days of training and matches.	Daily / weekly planning. Most actionable view — "who is overloaded RIGHT NOW?"
14 days	Two-week trend. Smooths out one heavy match week.	Mid-block reviews. Useful when MD-3/MD-2 falls in the middle of the week.
28 days	Monthly chronic profile. Closer to the chronic load reference Gabbett's papers used.	Mesocycle reviews. Spotting trends that the 7-day view misses.

> **Important about the slider**
> The 7/14/28 toggle changes the QUADRANT CHART aggregation only. The Squad Load table below the chart always shows 7-day acute / 28-day chronic / ACWR per metric — the slider doesn't apply to it.

6. The Squad Load table — Lite vs Full columns

Below the chart sits the Squad Load table. This is the per-player numerical companion to the visual chart above — every player gets a row, with 7-day acute, 28-day chronic, and ACWR per metric. The columns shown depend on your Catapult tier.

On Lite — Football metric set

Lite teams see metrics that work on every Catapult plan. The B2-3 efforts columns are deliberately swapped out for the highest-evidence injury proxies in the Lite-tier literature (Malone 2017 for HSR, Buchheit 2014 for max velocity).

Lite column	What it tracks
Total Distance (m)	Total ground covered. The volume baseline.
HSR Distance (m)	Distance run above ~19.8 km/h. Malone 2017 hamstring injury predictor.
Sprint Distance (m)	Distance run above ~25.2 km/h. Acute high-speed exposure.
Vel Band 5 Dist (m)	Distance in velocity band 5 (high-speed running zone).
Vel Band 6 Dist (m)	Distance in velocity band 6 (sprint zone).
Sprint Efforts (#)	Discrete sprint event count. Each sprint = one event.
Max Velocity (km/h)	Peak speed reached. Buchheit 2014 protective-dose anchor.

On Full (Pro / Premium) — Football metric set

Pro / Premium teams see the same volume + speed columns plus four B2-3 / Tot Accel-Decel columns that Lite plans don't expose:

- Accel B2-3 Efforts (Gen 2) — high-intensity acceleration count.
- Tot Accels (#) — total acceleration count across all bands.
- Decel B2-3 Efforts (Gen 2) — high-intensity deceleration count (the McBurnie 2022 input).
- Tot Decels (#) — total deceleration count across all bands.

What you DON'T see on Lite

The four columns above (Accel B2-3, Tot Accels, Decel B2-3, Tot Decels) do not appear in your Squad Load table — Lite plans don't expose those parameters in OpenField. The columns are not hidden because of a bug; they are deliberately removed because all values would be empty. The Lite metric set replaces them with HSR, Sprint, Sprint Efforts and Max Velocity — the highest-evidence Lite-equivalent injury proxies.

How the ACWR column reads

Every metric column shows three numbers per player: 7-day acute average, 28-day chronic average, and the acute:chronic ratio (ACWR). Apply the standard Gabbett interpretation:

ACWR	Interpretation
< 0.8	De-training paradox — chronic load is being lost. Vulnerable on return.
0.8 — 1.3	Sweet spot. Healthy progression.

ACWR	Interpretation
1.3 — 1.5	Watch — acute is starting to outpace chronic.
> 1.5	Spike — significantly elevated injury risk (Malone 2017: +200% for HSR).

7. Quick reference — Lite vs Full on this page

Page element	Lite tier	Pro / Premium tier
Citation badge	Gabbett 2016 · Volume axis (blue)	Gabbett 2017 · Full axis (green)
Quadrant chart axes	**FULL fidelity**	**FULL fidelity**
Quadrant counts bar	**FULL fidelity**	**FULL fidelity**
Time-window slider (7/14/28)	**Works**	**Works**
ACWR (dot size + colour)	**Works**	**Works**
Squad Load — Volume / HSR / Sprint columns	**Shown**	**Replaced by B2-3 columns**
Squad Load — B2-3 efforts columns	**Hidden**	**Shown**
EDI (Equivalent Distance Index)	Often null (needs HMLD, a Premium parameter)	Available with Premium HMLD parameter

8. Workflow — when to use Quadrant View

Daily — morning team-state read (2 min)

- Open the page on the way to morning planning. Default 7-day window is already what you want.
- Read the four quadrant counts first — that's your team-state in 5 seconds. Are people clustered in INJURY RISK? Pull back. Lots in PEAK FITNESS? You can push.
- Scan the chart for any RED dots — those are players with ACWR ≥ 1.5. Cross-check with HSR Intelligence and Foster Monotony & Strain.

Weekly — match-week planning (10 min)

- Switch to 14-day view to smooth out single heavy match weeks.
- Identify the DECOUPLED quadrant — these players need attention BEFORE they show up as injuries. Often the early signal of a fatigue / illness wave.
- Use the Squad Load table to see per-metric ACWR. A player can be GREEN on the quadrant (within team-median range) but RED on a specific metric like HSR or Sprint Distance — that's the deeper read.

Mesocycle review — monthly (15 min)

- Switch to 28-day view. This is closer to Gabbett's chronic-load reference.

- Look at how many players have been in INJURY RISK across consecutive weeks — chronic accumulation is what predicts injury, not single-week spikes.
- Compare to your injury log — were the players who got injured previously in INJURY RISK or DECOUPLED before the injury? That's how you build trust in the chart.

9. Frequently asked questions

Why does the same player move between quadrants from week to week?

Quadrant membership is RELATIVE to the team median — half the squad is always above the X-line, half below. If a player's absolute load doesn't change but their teammates train harder than usual, the player can shift from "high external" to "low external" without doing anything different. This is a feature, not a bug — it tells you who is doing more or less than the team average right now, which is the right question for daily planning.

Why is my player not on the chart at all?

The chart only shows players who have BOTH external load (Catapult data) AND internal cost (RPE entries) in the selected window. If a player is missing entirely, check both: are GPS sessions syncing for them? Are they logging RPE? A player on either axis but not both will not appear.

What if my team has very low RPE compliance?

The Y-axis (internal cost) needs RPE entries. Without them the chart will be sparse or empty. Fix RPE compliance before relying on this page — even if just one or two players log honestly, you can use them as a calibration reference. Foster Monotony & Strain has a volume-fallback variant; Quadrant View does not — RPE is required.

Should I trust the Lite chart as much as the Pro chart?

For the 2 × 2 quadrant view: yes. The chart axes are based on universal Catapult metrics (total distance + sRPE) — Lite and Pro produce identical chart fidelity. The Squad Load table below is what differs — Lite shows HSR / Sprint / Max Velocity columns instead of the Pro-only B2-3 efforts columns. Both metric sets give you valid injury proxies; B2-3 efforts are higher-fidelity for decel-mechanism injury (ACL / quad / patellar tendon), HSR / Sprint / Max V are higher-fidelity for sprint-mechanism injury (hamstring).

How does Quadrant View relate to HSR Intelligence and Foster Monotony & Strain?

Quadrant View = team-state visual at a glance. HSR Intelligence = per-player chronic injury risk from Malone 2017 + Buchheit 2014 + Bowen 2017. Foster Monotony & Strain = per-player overtraining risk from day-to-day variability. Same dataset, three different views. Use all three — the player who is RED on Quadrant + RED on HSR Intelligence + DANGER on Foster is your highest-priority case.

Why are the quadrant lines at the median and not at fixed thresholds?

Vanrenterghem and Gabbett's framework is designed for RELATIVE comparison within the team. A 5000 m / day average means very different things in a goalkeeper-heavy squad vs a midfielder-heavy squad. Team-median lines auto-adjust for your roster and your sport's typical loads. Absolute thresholds would have to be re-calibrated per sport and per club, which the framework deliberately avoids.

10. Limitations — what Quadrant View CAN'T do

- Requires both Catapult external load AND RPE entries. Players with only one are excluded.

- Quadrant lines are at the team median, so the chart is RELATIVE — a player can move between quadrants based on what teammates do, not just on their own training.
- Aggregates over the selected window — single high-intensity moments (e.g. a 30-min HIT block embedded in a long week) can be diluted by easier days.
- On Lite, the Squad Load table cannot show B2-3 efforts (decel-mechanism injury proxy). Use HSR Intelligence as the Lite-equivalent for that injury class.
- EDI (di Prampero 2015 Equivalent Distance Index) requires HMLD, typically a Premium parameter. Often null on Lite.
- Does not factor in subjective wellness (sleep, soreness, mood). Cross-check with Decision Summary wellness scores (Saw 2016).

11. References

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MicroPulse — Quadrant View (Lite) Coach Guide · v1.0 · May 2026